

Re-analysis Summary Report for Peer-evaluation

Paper ID: Christensen_EurJournPersonality_2018_8R9d

Re-analyst's ID: zxd19

Your task

As a peer-evaluator, for this paper you are asked to judge whether the analyst's applied analytical choices for Task 1 and Task 2 are acceptable. By acceptable, we mean that the analysis pipeline is within the variations that could be considered appropriate by the scientific community in addressing the given analytical task. In addition, for Task 1, we ask you to judge whether the reported conclusion adequately follows from the results of the analysis; whether the co-analyst's self-categorization of the result is adequate. Although, you are not required, it would be appreciated, if you could execute the accompanying analysis code to check any mismatch between the results of the analysis and the reported results.

For both tasks, you should provide your evaluation via this survey: <https://forms.gle/EAREaC6JjN2PWFfR6>

The re-analyst's task

The re-analyst was instructed to assess the following claim of the original authors:

... the high openness to experience group's network was more interconnected ... than the low openness to experience group (p. 480.)

The corresponding article

Here you can find the original article: https://osf.io/6ve3b/?view_only=c874fcf097644e82a320679d4641163a and the original data: https://osf.io/6zh34/?view_only=3e5bfed2e1f041dbbb4376e9c0c5f70b

They are provided only as context of the re-analysis.

In Task 1, the analyst was asked to conduct the analysis without any restrictions. Here, the analyst reported the following steps and results of the analysis:

Data was used that had been preprocessed following the procedures described in Christensen et al (2018). (I used the preprocessed "FINAL" data files.) Three analytic strategies were then used to test the claim of Christensen et al (2018).

First, I attempted to reproduce their results using the analytic approach described in Christensen et al (2018). Participants were split into two groups, one "high" in openness-to-experience and the other "low," on the basis of a median split of the summary measure of openness-to-experience described in Christensen et al (2018). Using only words that were produced by at least two participants in each group, I constructed separate semantic networks for each group by calculating the cosine similarity between co-occurrence vectors for each word. This word similarity matrix was further simplified by calculating the triangulated maximally filtered graph (TMFG) and then creating an unweighted graph (i.e., edges either exist or they do not). To test the main claim — that openness-to-experience is associated with shorter Average Shortest Path Length in semantic networks — I bootstrapped subnetworks that consisted of a random subset of 90% of all nodes ($n = 1000$ bootstrap iterations). These paired subnetworks were then compared. This analysis reproduced the results of Christensen et al: Overall, the ASPL was shorter for the high-openness group ($M = 3.1$) than

low-openness group ($M = 4.1$); $t_{999} = -77$, $p \ll .001$, $d = -2.7$. These numbers are similar to those reported in the original paper.

This approach, however, does not account for the possibility that a few unusual participants could completely bias the group-level estimates, since the analysis described above pools together all individuals within a group. I thus attempted two other approaches.

In the first alternative approach, I bootstrapped at the level of individual participants rather than nodes (i.e., words) ($n = 1000$ bootstrap iterations). Subsamples of participants ($n = 20$) were selected without replacement from the full sample. I then created a semantic network for this subgroup, following the same procedure as above, and calculated each subgroup's mean openness-to-experience and the APSL for their semantic network. According to this approach, there was a significant association between APSL and openness-to-experience ($p = .001$), but it was in the wrong direction: subgroups *higher* in openness-to-experience actually had *longer* APSL, Pearson's $r = 0.10$, $t_{998} = 3.3$. This was true even after controlling for the number of words in each bootstrapped semantic network, although this unpredicted effect was no longer statistically significant ($b = 0.08 \pm 0.09$ SE, $p = .4$). This approach to analyzing the data thus contradicted the approach described above.

In the second alternative approach, I calculated the average shortest path for groups of subjects created by binning the openness-to-experience measure. Bins of width 0.10 were created ($n = 12$), and for each bin I created a semantic network for individuals within that bin. There was no relation between mean openness-to-experience and APSL ($r = .49$, $p = .10$). A graph of the relationship showed a very erratic relationship between the two measures, and a numerical relationship in the unpredicted direction. This approach was thus inconclusive about any potential relationship between openness and APSL.

The analyst's conclusion: The data are inconclusive as to whether the High openness to experience group's semantic network has a lower average shortest path length (ASPL) than the Low openness to experience group's network.

The analyst's categorization of the result:

The results do not show evidence for or against the relationship/effect as described in the claim provided in your task

In Task 2, the analysts received the following instruction to conduct the analysis:

In the partial network that you use, the rate of nodes remaining should be 90%.

Here, the analyst reported the following steps and results of the analysis:

The analysis was identical to Task 1, except here I used only the first of the three approaches that I used for Task 1. I did this because the instructions explicitly stated: "In the partial network that you use, the rate of nodes remaining should be 90%." The other two approaches I used in Task 1 did not use the node sub-sampling approach that was adopted by the original others, so for this task I focused only on directly reproducing their analysis.

The results of this analysis were the same as in Task 1. Average shortest path length (ASPL) was shorter, on average, for subnetworks from high-openness participants ($M = 3.1$) than for subnetworks from low-openness participants ($M = 4.1$). This difference was statistically significant, paired Student's t-test, $t_{999} = -77$, $p \ll .0001$.

Analysis language/software/code (optional to check):

Task 1: R

Task 2: R

Analysis codes: <https://osf.io/2anzp/>